

On the Intrinsic and Extrinsic Fairness Evaluation Metrics for Contextualized Language Representations

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Abstract

Multiple metrics have been introduced to measure fairness in various natural language processing tasks. These metrics can be roughly categorized into two categories: 1) *extrinsic metrics* for evaluating fairness in downstream applications and 2) *intrinsic metrics* for estimating fairness in upstream contextualized language representation models. In this paper, we conduct an extensive correlation study between intrinsic and extrinsic metrics across bias notions using 19 contextualized language models. We find that intrinsic and extrinsic metrics do not necessarily correlate in their original setting, even when correcting for metric misalignments, noise in evaluation datasets, and confounding factors such as experiment configuration for extrinsic metrics.

1 Introduction

Recent natural language processing (NLP) systems use large language models as the backbone. These models are first pre-trained on unannotated text and then fine-tuned on downstream tasks. They have been shown to drastically improve the downstream task performance by transferring knowledge from large text corpora. However, several studies (Zhao et al., 2019; Barocas et al., 2017; Kurita et al., 2019) have shown that societal bias are also encoded in these language models and transferred to downstream applications. Therefore, quantifying the biases in contextualized language representations is essential for building trustworthy NLP technology.

To quantify these biases, various fairness metrics and datasets have been proposed. They can be roughly categorized into two categories: *extrinsic* and *intrinsic* metrics (Goldfarb-Tarrant et al., 2021). *Intrinsic fairness metrics* probe into the fairness of

the language models (Guo and Caliskan, 2021; Kurita et al., 2019; Nadeem et al., 2020; Nangia et al., 2020), whereas *extrinsic fairness metrics* evaluate the fairness of the whole system through downstream predictions (Dhamala et al., 2021; Jigsaw, 2019; De-Arteaga et al., 2019). Extrinsic metrics measure the fairness of system outputs, which are directly related to the downstream bias that affects end users. However, they only inform the fairness of the combined system components, whereas intrinsic metrics directly analyze the bias encoded in the contextualized language models.

Nevertheless, the relationship between upstream and downstream fairness is unclear. While some prior work has demonstrated that biases in the upstream language model have significant effects on the downstream task fairness (Jin et al., 2021), others have shown that intrinsic and extrinsic metrics are not correlated (Goldfarb-Tarrant et al., 2021). These studies either focus on one specific application or consider static word embeddings. Therefore, it is still obscure how fairness metrics correlate across different tasks that use contextualized language models.

To better understand the relationship between intrinsic and extrinsic fairness metrics, we conduct extensive experiments on 19 pre-trained language models (BERT, GPT-2, etc.). We delve into three kinds of biases, *toxicity*, *sentiment*, and *stereotype*, with six fairness metrics across intrinsic and extrinsic metrics, in text classification and generation downstream settings. The protected group domains we focus on are *gender*, *race*, and *religion*.

Similar to the observations in static embeddings (Goldfarb-Tarrant et al., 2021), we find that these metrics correlate poorly. Therefore, when evaluating model fairness, researchers and practitioners should be careful in using intrinsic metrics as a proxy for evaluating the potential for downstream biases, since doing so may lead to failure to detect bias that may appear during inference. Specifi-

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cally, we find that correlations between intrinsic and extrinsic metrics are sensitive to alignment in notions of bias, quality of testing data, and protected groups. We also find that extrinsic metrics are sensitive to variations on experiment configurations, such as to classifiers used in computing evaluation metrics. Practitioners thus should ensure that evaluation datasets correctly probe for the notions of bias being measured. Additionally, models used to compute evaluation metrics such as those in BOLD (Dhamala et al., 2021) can introduce additional bias, and thus should be optimized to be robust.

The main contribution of our work is as follows: First, we conduct an extensive study on correlations between intrinsic and extrinsic metrics. Second, we conduct ablation studies to show the effect of (mis)alignment of notions of bias and protected groups, and noise in recent fairness evaluation datasets. Finally, we conduct a study on the robustness (or lack thereof) of sentiment classifiers to text generation quality, diversity, and noise.

With this paper, we aim to suggest preliminary best practices in bias evaluation, which is important to quantify progress in debiasing work that will make models usable in the real world.

2 Fairness Evaluation Metrics

The notion of bias we focus on is group disparities in *sentiment*, *toxicity* and *stereotypes*, which have been extensively researched in the context of fairness (Huang et al., 2020; Zhang et al., 2020; Borkan et al., 2019; Adragna et al., 2020). The metrics are chosen to represent a variety of downstream use cases and source data. We use CEAT and ILPS for intrinsic metrics for toxicity and sentiment, HATEX, TOXD, and B-TOX for extrinsic toxicity metric, and B-SENT for extrinsic sentiment metric. For stereotype, we use STEREOSET and B-STEREO for intrinsic and extrinsic metrics.

Intrinsic: Contextualized Embedding Association Test (CEAT) (Guo and Caliskan, 2021) measures bias through word embedding distances in semantic vector space. Given a set of protected group (e.g. “woman”) and two polar attribute sets (e.g. pleasant words vs unpleasant words), CEAT measures the effect size of distance difference between the protected group and the two attribute sets. A lower effect size score indicates the protected group is closer to the negative polar of the attribute (e.g. *unpleasant*) and thus probably more

correlated with toxic words.

Intrinsic: Increased Log Probability Score (ILPS) (Kurita et al., 2019) measures bias on word associations between protected groups and attributes with templates such as “The [group] is <mask>.” It computes the increase in probability of the attribute (e.g. “happy”) for the <mask> token given a protected group (e.g. “woman”). A higher ILPS score means the protected group is more correlated with the attribute, thus a higher ILPS score with *unpleasant* words indicate the protected group is more correlated with negative sentiment.

Intrinsic: StereoSet (Nadeem et al., 2020) measures bias in language model scoring of stereotyped sentences over anti-stereotyped sentences. Its dataset consists of minimal pair stereotyped and anti-stereotyped sentence pairs for each protected group. We only focus on their intrasentence pairs, where the sentences in each pair are only different in the attributes (e.g. “The Iranian man might be a *terrorist* individual” and “The Iranian man might be a *hardworking* individual” is a sentence pair for Iranian group). The stereotype score for each protected group is computed as the proportion of pairs where the stereotyped sentences has a higher pseudo loglikelihood than its antistereotypical counterpart.

Extrinsic: Jigsaw Toxicity (TOXD) (Jigsaw, 2019) measures bias in toxicity detection systems that covers multiple protected groups. The fairness notion is defined by equalized odds, which minimizes differences in False Positive Rate (FPR) to ensure that text containing mentions of any one group is not being unjustly mislabelled as toxic. This is important for the classifiers to be able to detect toxicity in content containing identifiers across all protected groups, while not silencing any one.

Extrinsic: HateXPlain (HATEX) (Mathew et al., 2020) measures bias in hate speech detection systems. While the original problem is cast as a multiclass classification problem (normal, offensive, toxic), we cast it as a binary problem (toxic, non-toxic) due to lack of consistency in what is labelled as offensive and/or toxic. Similar to TOXD, the measure of bias against a certain group is the False Positive Rate on examples with group mentions.

Extrinsic: BOLD (Dhamala et al., 2021) is a dataset that measures bias in language generation that consist of Wikipedia-sourced natural prompts. Given a prompt containing direct or indirect mentions of a protected group, BOLD evaluates the

quality of the sentences finished by the language model. We focus on the sentiment (B-SENT) metric for sentiment, toxicity (B-TOX) metric for toxicity, and regard (B-REGARD) metric for stereotype. Additionally, for stereotype, we train a stereotype classifier by finetuning the BERT model with StereoSet (Nadeem et al., 2020), CrowS-Pairs (Nangia et al., 2020), and Social Bias Frames (Sap et al., 2020) datasets, and use this classifier to evaluate BOLD generations on stereotype (B-STEREO)³.

The bias score for each protected group is calculated as the average toxicity, sentiment, regard, and stereotype score on the generations from the prompts with that protected group.

3 Correlation between Metrics

Experiment Setup We conduct a study on *gender*, *race*, and *religion* domains (see the Appendix A for the list of protected groups on each domain). We conduct correlation analysis on the **variance** of group metric scores across protected groups, as it captures score disparities across protected groups for each domain. For example, for $M = \text{CEAT}$, we define $S_{M_{\text{race}}} = \text{Var}(s_{\text{Asian}}, s_{\text{White}}, s_{\text{Black}}, \dots)$. A less-biased model would have smaller variance score. Thus, if two metrics are correlated, we would see a positive correlation, as reducing the disparity between groups in one metric, as measured by variance would reduce that in the other.

We evaluate 19 popular pre-trained language models⁴. These models consist of ALBERT (Lan et al., 2020) (base-v2, large-v2, xlarge-v2, xxlarge-v2), BERT (Devlin et al., 2019) (base-cased, large-cased), RoBERTa (base, large), DistilRoBERTa (Sanh et al., 2019), GPT2 (Radford et al., 2019) (base, medium, large, xl), DistilGPT2, EleutherAI/gpt-neo (Black et al., 2021) (125M, 1.3B, 2.7B), and XLNet (Yang et al., 2019) (base-cased, large-cased)⁵. For intrinsic metrics, we simply measure the corresponding metric scores on the language models⁶. For extrinsic metrics, we

³The stereotype classifier reaches a F1 score of 0.80 on the validation dataset. See Appendix B for training details.

⁴Code are available at <https://github.com/pruksmh/fairness-metrics-correlations>

⁵We pick the most popular models for both masked language models and generative language models from Huggingface <https://huggingface.co/models>.

⁶We use the same experiment settings, such as testing word choices, testing dataset, etc., as proposed in the papers where these metrics are introduced. CEAT does not cover groups in *religion*, so we adopt the protected group list from Sotnikov

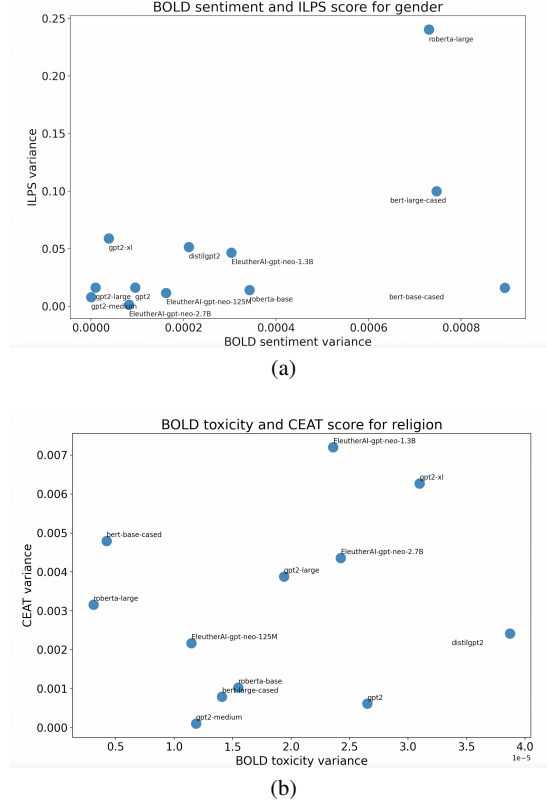


Figure 1: Examples of the correlation plots on ILPS versus B-SENT (a) and CEAT versus B-Tox (b). Each point represents a language model.

	Gender		Race		Religion	
	CEAT	ILPS	CEAT	ILPS	CEAT	ILPS
ToxD	-0.12	0.26	-0.06	-0.37	0.28	-0.37
HATEX	-0.12	0.10	-0.05	0.73	0.23	-0.38
B-TOX	0.21	-0.28	0.41	-0.34	0.19	-0.53
B-SENT	-0.03	0.54	0.67	0.30	-0.42	-0.58

Table 1: Correlation results on toxicity and sentiment metrics. Results in bold are statistically significant.

fine-tune language models for classification-based tasks⁷, and either sample in an autoregressive manner for autoregressive language models, or use random masking-based generation for MLM-based models (Wang and Cho, 2019) following the BOLD paper, for generation-based tasks⁸.

For each intrinsic and extrinsic metric pair, we take the intrinsic and extrinsic scores for each et al. (2021) for the *religion* domain.

⁷For classification tasks, we use the hyper-parameters for fine-tuning language models on tasks for extrinsic metrics that achieve close to state-of-the-art F1 score (see Appendix C).

⁸Some language models are not suitable for the generation task due to the nature of the language model’s pre-training method. Thus we exclude these models, including ALBERT, DistilRoBERTa, and XLNet models, for BOLD-related extrinsic metrics calculation.

	STEREOSet		
	Gender	Race	Religion
B-STEREO	-0.32	-0.18	0.10
B-REGARD	-0.21	-0.08	-

Table 2: Correlation results on stereotype metrics. The regard classifier is not trained with any data on religion. Thus we do not apply it to the BOLD generations for religion.

	Gender	Race	Religion
	CEAT _{TOX}	CEAT _{TOX}	CEAT _{TOX}
TOXD	0.04	0.08	0.42
HATEX	0.17	0.49	0.43
B-TOX	0.91	0.41	0.56
B-SENT	-0.46	-0.18	0.38

Table 3: Correlation results between CEAT_{TOX} and toxicity extrinsic metrics. Results in bold are statistically significant.

model. With the list of score pairs from the 19 models, we compute the correlation using the Pearson correlation coefficient. If the metrics are positively correlated, the correlation score should be close to 1. Figure 1 depicts some examples of the correlation plots.

Correlation Results Table 1 contains correlations scores for each intrinsic/extrinsic metric pair on sentiment and toxicity. Only few metrics have significantly positive correlations. In general, ILPS has more significantly positive correlations with the extrinsic metrics compared to CEAT, except for the religion domain. This may due to the nature of the two intrinsic metrics – ILPS is calculated with log probabilities, which is more related to the downstream generative tasks such as BOLD since generation samples based on log probabilities.

For sentiment metrics, we find more statistically significant positive correlations between intrinsic metrics and B-SENT than toxicity extrinsic metrics.

In both toxicity and sentiment, we see that there are statistically negative correlations for the religion domain, which we investigate in Section 3.2.

For stereotype, Table 2 contains the results on stereotype metrics. We see that none of the correlations are significant nor positive.

4 Ablation Study

There are many factors at play in fairness evaluation processes, such as notion of bias measured, choice of protected groups, quality of the testing data, and confounding factors in the models used to compute metrics themselves. In this section, we conduct careful analysis to explore why extrinsic and intrinsic metrics are not always correlated.

4.1 Misalignment between metrics

In our main study, we use the experimental settings defined in their original papers. However, these metrics may have subtle misalignments in type of bias measured, protected groups factored in calculation, and characteristics of the evaluation dataset.

Misalignment on the notion of bias Among the toxicity metrics, the notion of bias are not consistent – some measure sentiment (CEAT, ILPS, B-SENT) while others measure toxicity. Therefore, we recompute CEAT scores with toxicity word seeds, which we denote as CEAT_{TOX}. We manually pick 20 *toxic* and 20 *anti-toxic* words from the word clouds of the toxic and non-toxic labeled sentences in the JigsawToxicity dataset for CEAT_{TOX}. See Appendix D for the full list of the words.

As seen in Table 3, the correlations between the toxicity-related extrinsic metrics and CEAT_{TOX} are more positive than with CEAT. Also note that CEAT is better correlated with B-SENT than CEAT_{TOX}, except for religion. Though many of the correlation scores remain not statistically significant, the result supports our hypothesis that intrinsic and extrinsic metrics are more correlated when they have the same notion of bias.

Misalignment on the protected groups Due to the limited number of overlapping protected groups (stereotype metrics only have four groups in common), we compute the domain-level variance scores for all protected groups contained in a dataset. However, the groups that are not present in both the evaluation datasets for intrinsic and extrinsic metrics may introduce metric disalignment, as they would be factored in metric computation in one but not the other. We recompute the correlation of STEREOSet with B-REGARD and B-STEREO with only overlapping protected race groups⁹: White, Black, Hispanic, and Asian.

We find the correlation of STEREOSet with B-REGARD raises from -0.08 to 0.19 (p-value 0.56). The correlation with B-STEREO increases from -0.18 to 0.08 (p-value 0.80). These metrics are more positively correlated with the aligned groups.

Misalignment on evaluation dataset We observe that dataset sources for certain metrics are misaligned, such as that for BOLD and STEREOSet. STEREOSet uses crowdworkers to generate testing data specifically to contain particular

⁹STEREOSet does not have group Asian and White, so we use Japanese and Britain instead for these groups.

stereotypes. On the other hand, BOLD prompts are sourced from Wikipedia, which consist of more formal writing and is not directly engineered to probe for stereotypes. Examples of source misalignment can be seen in the Appendix.

To align the stereotype metrics, we use data from the STEREOSET intersentence dataset, which consists of a one-sentence context followed by a relevant stereotyped sentence, to compute BOLD metrics. Specifically, we use the context sentence for BOLD-like generation (see Appendix F for generation examples). We test STEREOSET with the new B-STEREO on the *race* domain and find that the correlation score increase from -0.18 to 0.02 (p-value 0.98). This indicates that aligning the evaluation dataset source has a modest impact on improving correlation between metrics.

4.2 Noise in Evaluation Datasets

As pointed out in Blodgett et al. (2021), some fairness evaluation datasets lack consistency in framing and data collection methodology, which leads to datasets not properly evaluating the intended notion of bias. We find evidence of this phenomena in the BOLD dataset for religion prompts, which contain toxic and stereotyped content, which will bias generations to be more toxic for certain groups. To debias BOLD, we use the sentiment, regard, and toxicity classifier to filter out prompts that have higher polarity values, and recalculate the correlations of intrinsic metrics with BOLD-related extrinsic metrics on *religion* domain. We find that scores for CEAT and B-SENT increases to 0.11, STEREOSET and B-STEREO increases to 0.10. This indicates that bias in datasets can affect the metrics.

4.3 Effect of Experiment Configuration on Metric Scores

Experiment configurations may also affect the amount of bias detected in fairness metrics, which we observe in BOLD metrics. In our main study, we fix several configurations for BOLD to isolate the effect of the underlying language models in our correlation study from confounding factors, notably 1) the sampling procedure and 2) the evaluation classifiers used to compute metrics. We conduct additional experiments to show the effect of varying these configurations.

Impact of sampling temperature on classifier-based metrics We input five sample prompts (listed in Appendix G) from BOLD dataset to GPT-2

model and for each prompt, generate 100 sentences. We use two temperature settings ($T = 0.5$ and $T = 1.0$) and compute the average sentiment over the generated sentences. We observe that the proportion of negative sentiment assignment increases from 4.6% to 15.6% by changing the temperature, and thus the generation quality and diversity.

Impact of noise in generated outputs on classifier based metrics We introduce noise to 500 BOLD generations through word swaps or deletions (examples shown in Appendix H)¹⁰. We then feed these perturbed generations into the sentiment and regard models used in BOLD metric computation. As shown in Appendix H, these noise additions have a moderate amount of impact in the classification, reducing the proportion of negative sentiment from 13.6% to 12.18% and proportion of negative sentiment from 25.2% to 22.86%.

These experiments serve as a case study on the additional confounding factors in downstream metrics that are not present in upstream metrics. Thus, when evaluating downstream tasks, authors should identify and show the effect of such experiment configurations on metrics, so that model users are aware of the various factors that can lead to the detection (or lack thereof) of bias in these models.

5 Conclusion

We present a study on intrinsic and extrinsic fairness metrics in contextualized word embeddings. Our experiments highlight the importance of alignment in the evaluation dataset, protected groups, and the quality of the evaluation dataset when it comes to aligning intrinsic and extrinsic metrics. Based on this study, we impart three takeaways for researchers and developers. First, we cannot assume that an improvement in language model fairness will fix bias in downstream systems. Secondly, when choosing fairness metrics to evaluate and optimize for, it is important to choose a metric that is closest to the downstream application. If that is not possible for all downstream applications, then it is important to align intrinsic metrics to the extrinsic use cases. Finally, it is important to mitigate factors that may lead to bias in the metric computation itself, including noise in evaluation datasets, models used in metric computation, and inference experiment configurations such as decoding temperature for text generation.

¹⁰The noise in this dataset may not reflect that in the real world.